PROBABILITY AND STATISTICAL MODELING FOR COMPUTING SYSTEMS

Question & Answer Bank – Units I to V
Part A: Multiple-Choice Questions (1 Mark)

KPR Institute of Engineering and Technology | Department of Mathematics | Semester III | R2025

Instructions: Choose the correct option. The answer and a short explanation are provided immediately after each question. Questions follow the topics and Part A structure of the uploaded question bank.

UNIT I – PROBABILITY AND DISCRETE RANDOM VARIABLES

20 Multiple-Choice Questions • Total: 20 Marks

1. The probability of throwing an even number on a fair die is:

A. 1/6
B. 1/2
C. 1/3
D. 1/4

Answer: B. 1/2 — There are 3 even outcomes among 6.

2. The probability of an impossible event is:

A. 1
B. 0
C. 1/2
D. 1/4

Answer: B. 0 — An impossible event has probability zero.

3. From 52 cards, the probability of drawing a black card or a queen is:

A. 1/13
B. 13/26
C. 7/13
D. 2/13

Answer: C. 7/13 — (26 + 4 – 2)/52 = 28/52 = 7/13.

4. If A and B are independent, P(A)=0.4 and P(B)=0.5, then P(A∪B) is:

A. 0.7
B. 0.9
C. 0.2
D. 0.3

Answer: A. 0.7 — $P(A \cup B)=P(A)+P(B)-P(A)P(B)=0.7$.

5. For a sample space S:

A. P(S)=1
B. P(S)=0
C. P(S)<1
D. P(S)>1

Answer: A. P(S)=1 — The probability of the entire sample space is 1.

6. The conditional probability of B given A is:

A. $P(A \cap B)/P(A)$
B. $P(A)/P(B)$
C. $P(A \cup B)$
D. $P(A)P(B)$

Answer: A. $P(A \cap B)/P(A)$ — $P(B|A)=P(A \cap B)/P(A)$, provided $P(A)>0$.

7. A random variable that assumes a countable number of values is:

A. Continuous
B. Discrete
C. Density function
D. None

Answer: B. Discrete — A discrete random variable has countable possible values.

8. A probability mass function must satisfy:

A. $p(x)<0$
B. $\Sigma p(x)=1$ and $p(x) \geq 0$
C. $\Sigma p(x)=0$
D. $p(x)>1$

Answer: B. $\Sigma p(x)=1$ and $p(x) \geq 0$ — All probabilities are non-negative and sum to one.

9. The process whose outcome is not predictable with certainty is a:

A. Random experiment
B. Sample space
C. Constant
D. Parameter

Answer: A. Random experiment — A random experiment can produce different outcomes.

10. Var(aX+b) is:

A. $aVar(X)+b$
B. $a^2Var(X)$
C. $a^2Var(X)+b$
D. $Var(X)+b$

Answer: B. $a^2Var(X)$ — Adding a constant does not change variance; scaling squares the factor.

11. E(aX+b) is:

A. $aE(X)+b$
B. $a^2E(X)$
C. $E(X)+ab$
D. $a+E(X)$

Answer: A. $aE(X)+b$ — Expectation is linear.

12. If A and B are mutually exclusive, then:

A. $P(A \cap B)=0$
B. $P(A \cap B)=1$
C. $P(A)=P(B)$
D. They are independent always

Answer: A. $P(A \cap B)=0$ — Mutually exclusive events cannot occur together.

13. The probability of obtaining a sum of 7 when two fair dice are rolled is:

A. 1/12
B. 1/6
C. 1/36
D. 1/3

Answer: B. 1/6 — Six of 36 equally likely outcomes give a sum of 7.

14. For a binomial distribution, the mean is:

A. np
B. nq
C. p/q
D. n/p

Answer: A. np — Mean of Binomial(n,p) is np.

15. If n=8 and p=0.5, the binomial mean is:

A. 2
B. 6
C. 4
D. 8

Answer: C. 4 — np=8×0.5=4.

16. For a Poisson distribution, the mean is denoted by:

A. λ
B. σ
C. p
D. q

Answer: A. λ — The Poisson parameter λ is both mean and variance.

17. For a Poisson distribution, mean and variance are:

A. Equal
B. Not equal
C. Always zero
D. Negative

Answer: A. Equal — Both are λ.

18. The addition law of probability is:

A. $P(A \cup B)=P(A)+P(B)-P(A \cap B)$
B. $P(A \cup B)=P(A)P(B)$
C. $P(A \cap B)=P(A)+P(B)$
D. $P(A)=1-P(B)$

Answer: A. $P(A \cup B)=P(A)+P(B)-P(A \cap B)$ — This is the general addition rule.

19. Bayes' theorem is used to find:

A. A posterior probability
B. Only the sample mean
C. The variance only
D. A time trend

Answer: A. A posterior probability — It updates a probability using observed evidence.

20. For Binomial(n,p), the variance is:

A. np
B. npq
C. nq
D. p^2

Answer: B. npq — Variance is npq, where q=1−p.

UNIT II – CONTINUOUS RANDOM VARIABLES

20 Multiple-Choice Questions • Total: 20 Marks

1. A continuous random variable can assume:

A. Only integer values
B. Only positive values
C. Any value within an interval
D. Only finite values

Answer: C. Any value within an interval — A continuous variable can take any value in an interval.

2. A probability density function is used for:

A. Discrete variables
B. Continuous variables
C. Only Bernoulli variables
D. Only binomial variables

Answer: B. Continuous variables — A PDF describes continuous random variables.

3. The total area under a valid probability density curve is:

A. 0
B. 0.5
C. 1
D. Infinity

Answer: C. 1 — The integral of a PDF over its support is one.

4. The CDF of a continuous random variable is:

A. Non-decreasing
B. Always decreasing
C. Always negative
D. Constant

Answer: A. Non-decreasing — A CDF is non-decreasing and ranges from 0 to 1.

5. For a continuous random variable, P(X=a) is:

A. 1
B. 0
C. 0.5
D. Depends on a

Answer: B. 0 — A single point has zero probability for a continuous variable.

6. The CDF is related to the PDF by:

A. $F(x)=\int$ from $-\infty$ to x $f(t)dt$
B. $F(x)=f'(x)$
C. $F(x)=1/f(x)$
D. $F(x)=f(x)+1$

Answer: A. $F(x)=\int$ from $-\infty$ to x $f(t)dt$ — The CDF is the accumulated area under the PDF.

7. The PDF is obtained from the CDF by:

A. Integration
B. Differentiation
C. Multiplication
D. Division

Answer: B. Differentiation — For a continuous variable, $f(x)=F'(x)$ where differentiable.

8. The mean of a continuous random variable is:

A. $\int xf(x)dx$
B. $\int f(x)dx$
C. $\int x^2 f(x)dx$
D. $1/F(x)$

Answer: A. $\int xf(x)dx$ — Integrate x times the density over its support.

9. The variance of X is:

A. $E(X^2)-[E(X)]^2$
B. $E(X)+E(X^2)$
C. $[E(X)]^2-E(X^2)$
D. $E(X^2)/E(X)$

Answer: A. $E(X^2)-[E(X)]^2$ — This is the standard variance identity.

10. A uniform distribution is also called:

A. Rectangular distribution
B. Bell-shaped distribution
C. Skewed distribution
D. Poisson distribution

Answer: A. Rectangular distribution — Its density is constant over the interval.

11. For X uniform on [a,b], the PDF inside the interval is:

A. $1/(b-a)$
B. $b-a$
C. $a+b$
D. $1/(a+b)$

Answer: A. $1/(b-a)$ — The constant density makes total area one.

12. For a uniform distribution on [2,8], the mean is:

A. 3
B. 4
C. 5
D. 6

Answer: C. 5 — Mean=(2+8)/2=5.

13. For X uniform on [a,b], the variance is:

A. $(b-a)^2/12$
B. $(a+b)/2$
C. $(b-a)/2$
D. $(a+b)^2/12$

Answer: A. $(b-a)^2/12$ — This is the uniform-distribution variance.

14. The graph of a normal distribution is:

A. Rectangular
B. Bell-shaped
C. Triangular
D. Exponential

Answer: B. Bell-shaped — The normal curve is symmetric and bell-shaped.

15. In a normal distribution, mean, median and mode are:

A. Different
B. Equal
C. Mean greater than median
D. Mode greater than mean

Answer: B. Equal — All three coincide at the centre.

16. The standard normal distribution has:

A. Mean 0 and SD 1
B. Mean 1 and SD 0
C. Mean 0 and variance 0
D. Mean 1 and SD 1

Answer: A. Mean 0 and SD 1 — Z is standard normal when $\mu=0$ and $\sigma=1$.

17. The standard score is calculated by:

A. $Z=(X-\mu)/\sigma$
B. $Z=(X+\mu)/\sigma$
C. $Z=\sigma/(X-\mu)$
D. $Z=X\mu\sigma$

Answer: A. $Z=(X-\mu)/\sigma$ — It measures distance from the mean in standard deviations.

18. For a normal distribution, the total area under the curve is:

A. 0
B. 0.5
C. 1
D. 2

Answer: C. 1 — The normal PDF integrates to one.

19. A uniform distribution on [3,15] has mean:

A. 6
B. 9
C. 12
D. 15

Answer: B. 9 — $(3+15)/2=9$.

20. A normal distribution is:

A. Symmetric about its mean
B. Always positively skewed
C. Always discrete
D. Uniform

Answer: A. Symmetric about its mean — The normal curve is symmetric.

UNIT III – TWO-DIMENSIONAL RANDOM VARIABLES, COVARIANCE AND REGRESSION

20 Multiple-Choice Questions • Total: 20 Marks

1. A joint PMF of two discrete random variables must satisfy:

A. ΣΣp(x,y)=0
B. ΣΣp(x,y)=1
C. p(x,y) may be negative
D. ΣΣp(x,y)=0.5

Answer: B. ΣΣp(x,y)=1 — All joint probabilities sum to one.

2. If X and Y are independent, their joint PMF is:

A. pX(x)pY(y)
B. pX(x)+pY(y)
C. pX(x)/pY(y)
D. pX(x)−pY(y)

Answer: A. pX(x)pY(y) — Independence means the joint distribution factors.

3. A joint PDF must satisfy:

A. ∫∫f(x,y)dxdy=1
B. f(x,y)<0 somewhere
C. ∫∫f(x,y)dxdy=0
D. f(x,y)=1 everywhere

Answer: A. ∫∫f(x,y)dxdy=1 — The total integral over the support is one.

4. Two continuous random variables are independent if:

A. f(x,y)=fX(x)fY(y)
B. f(x,y)=fX(x)+fY(y)
C. fX(x)=fY(y)
D. Cov(X,Y)=1

Answer: A. f(x,y)=fX(x)fY(y) — Factorization of the joint density characterizes independence.

5. Covariance is defined as:

A. E[(X−μX)(Y−μY)]
B. E(X)+E(Y)
C. Var(X)+Var(Y)
D. E(XY)+E(X)E(Y)

Answer: A. E[(X−μX)(Y−μY)] — Covariance measures joint variation.

6. If X and Y are independent with finite variances, Cov(X,Y) is:

A. 0
B. −1
C. 1
D. Undefined

Answer: A. 0 — Independence implies zero covariance.

7. The correlation coefficient is:

A. Cov(X,Y)/(σXσY)
B. Cov(X,Y)×σXσY
C. Var(X)/Var(Y)
D. E(XY)

Answer: A. Cov(X,Y)/(σXσY) — Correlation is standardized covariance.

8. The correlation coefficient lies in:

A. [–1,1]
B. [0,1]
C. [1,2]
D. (–∞,∞)

Answer: A. [–1,1] — Its magnitude cannot exceed one.

9. If the correlation coefficient is zero, the variables are:

A. Uncorrelated
B. Perfectly correlated
C. Always independent
D. Perfectly negative

Answer: A. Uncorrelated — Zero correlation means no linear correlation; not necessarily independence.

10. A correlation coefficient of –1 indicates:

A. Perfect negative linear relationship
B. Perfect positive linear relationship
C. No linear relationship
D. Independence

Answer: A. Perfect negative linear relationship — The variables lie exactly on a downward-sloping line.

11. A correlation coefficient of +1 indicates:

A. Perfect positive linear relationship
B. Perfect negative linear relationship
C. No relationship
D. Independence

Answer: A. Perfect positive linear relationship — The variables lie exactly on an upward-sloping line.

12. Covariance of X with itself is:

A. Var(X)
B. 0
C. 1
D. –Var(X)

Answer: A. Var(X) — Cov(X,X)=Var(X).

13. The marginal distribution of X is obtained from a joint discrete PMF by:

A. Summing over all y
B. Differentiating over y
C. Multiplying by y
D. Dividing by x

Answer: A. Summing over all y — Sum the joint probabilities over Y.

14. For independent X and Y, E(XY) equals:

A. E(X)E(Y)
B. E(X)+E(Y)
C. E(X)–E(Y)
D. 0 always

Answer: A. E(X)E(Y) — Expectations factor under independence.

15. The conditional PMF of X given Y=y is:

A. p(x,y)/pY(y)
B. p(x,y)pY(y)
C. pX(x)+pY(y)
D. pX(x)/p(x,y)

Answer: A. p(x,y)/pY(y) — Divide the joint probability by the marginal probability of Y=y.

16. Marginal distributions describe:

A. Each variable separately
B. Only covariance
C. Only correlation
D. Only joint behaviour

Answer: A. Each variable separately — They give the individual distributions.

17. The regression line of Y on X is used to:

A. Predict Y from X
B. Find Var(X) only
C. Find the median
D. Test a PDF

Answer: A. Predict Y from X — It estimates Y for a given X.

18. The correlation coefficient is independent of:

A. Change of origin and positive scale
B. All data values
C. The sample size
D. The means

Answer: A. Change of origin and positive scale — Correlation is invariant under changes of origin and positive scale.

19. If Cov(X,Y)>0, the variables tend to:

A. Move in the same direction
B. Move in opposite directions
C. Have no association of any kind
D. Be independent

Answer: A. Move in the same direction — Positive covariance indicates same-direction linear tendency.

20. The regression coefficients b_yx and b_xy satisfy:

A. $r^2=b_{yx}\, b_{xy}$
B. $r=b_{yx}+b_{xy}$
C. $r^2=b_{yx}/b_{xy}$
D. r=0 always

Answer: A. $r^2=b_{yx}\, b_{xy}$ — The product of regression coefficients equals the square of correlation.

UNIT IV – ESTIMATION AND HYPOTHESIS TESTING

20 Multiple-Choice Questions • Total: 20 Marks

1. Statistical inference uses _____ to draw conclusions about a population:

A. Population data
B. Sample data
C. Census only
D. Theoretical data

Answer: B. Sample data — Inference generalizes from a sample.

2. A portion of a population examined to determine its characteristics is called a:

A. Parameter
B. Statistic
C. Sample
D. Census

Answer: C. Sample — A sample is a subset of the population.

3. A numerical measure describing a population is a:

A. Statistic
B. Parameter
C. Estimator
D. Observation

Answer: B. Parameter — A parameter describes the population.

4. A numerical measure calculated from sample data is a:

A. Parameter
B. Statistic
C. Population
D. Census

Answer: B. Statistic — A statistic is computed from a sample.

5. An estimate expressed as a single value is called:

A. Interval estimation
B. Point estimation
C. Biased estimation
D. Sampling frame

Answer: B. Point estimation — Point estimation gives one value.

6. An estimate expressed as a range is called:

A. Point estimation
B. Interval estimation
C. Type I error
D. Test statistic

Answer: B. Interval estimation — An interval estimate gives a range of plausible values.

7. An estimator is unbiased if:

A. E(estimator)=parameter
B. Variance is zero
C. It is always positive
D. It equals the sample size

Answer: A. E(estimator)=parameter — Its expected value equals the parameter.

8. The sample variance with denominator n−1 is commonly used as an unbiased estimator of:

A. Population variance
B. Population mean
C. Population proportion
D. Sample size

Answer: A. Population variance — The corrected sample variance estimates σ^2 without bias.

9. A more efficient estimator generally has:

A. Smaller variance
B. Larger bias
C. Larger sample size always
D. Zero probability

Answer: A. Smaller variance — Among unbiased estimators, lower variance means greater efficiency.

10. A consistent estimator converges to the true parameter as:

A. Sample size increases
B. Sample size decreases
C. Time becomes zero
D. Variance increases

Answer: A. Sample size increases — Consistency is an asymptotic property.

11. The null hypothesis is denoted by:

A. H■
B. H■
C. α
D. β

Answer: A. H■ — H■ is the statement tested initially.

12. The alternative hypothesis is denoted by:

A. H■
B. H■
C. σ
D. n

Answer: B. H■ — H■ represents the competing claim.

13. Rejecting a true null hypothesis is called:

A. Type I error
B. Type II error
C. Sampling error
D. Standard error

Answer: A. Type I error — Type I error has probability α.

14. Failing to reject a false null hypothesis is called:

A. Type I error
B. Type II error
C. Level of confidence
D. Point estimate

Answer: B. Type II error — Type II error has probability β.

15. The probability of a Type I error is the:

A. Power
B. Level of significance
C. Confidence coefficient
D. Standard deviation

Answer: B. Level of significance — The significance level is α.

16. For a large-sample test of a mean with known population SD, the usual statistic is:

A. Z-statistic
B. t-statistic only
C. F-statistic
D. Chi-square only

Answer: A. Z-statistic — The standardized sample mean follows the Z approach.

17. At 95% confidence, the two-sided standard normal critical value is approximately:

A. 1.645
B. 1.96
C. 2.33
D. 2.58

Answer: B. 1.96 — The central 95% interval uses ±1.96.

18. The rejection area of a hypothesis test is called the:

A. Acceptance region
B. Critical region
C. Confidence interval
D. Sample space

Answer: B. Critical region — The critical region contains values leading to rejection.

19. For a contingency table with r rows and c columns, degrees of freedom are:

A. (r–1)(c–1)
B. r+c
C. rc–1
D. r–c

Answer: A. (r–1)(c–1) — This is the chi-square independence-test degree of freedom.

20. The chi-square test of independence examines whether two attributes are:

A. Independent
B. Normally distributed
C. Equal in mean
D. Continuous

Answer: A. Independent — It tests association between categorical attributes.

UNIT V – TIME SERIES ANALYSIS AND FORECASTING

20 Multiple-Choice Questions • Total: 20 Marks

1. The primary objective of time series analysis is to:

A. Classify observations
B. Forecast future values
C. Test hypotheses
D. Compare populations

Answer: B. Forecast future values — Time series analysis studies patterns over time for forecasting.

2. The long-term movement of a time series is called:

A. Seasonal variation
B. Cyclical variation
C. Secular trend
D. Irregular variation

Answer: C. Secular trend — Trend is the long-term direction.

3. Business expansion and recession over several years are examples of:

A. Seasonal variation
B. Cyclical variation
C. Trend only
D. Random sampling

Answer: B. Cyclical variation — Cyclical variation occurs over multi-year business cycles.

4. Time series forecasting helps managers to:

A. Remove all uncertainty
B. Support planning and decision-making
C. Eliminate all risks
D. Increase sample size

Answer: B. Support planning and decision-making — Forecasts support planning but do not eliminate uncertainty.

5. The least squares method is used to obtain the:

A. Seasonal index
B. Best-fitting trend line
C. Moving average only
D. Correlation coefficient

Answer: B. Best-fitting trend line — Least squares estimates trend coefficients.

6. In the linear trend equation ■=a+bt, b represents the:

A. Intercept
B. Mean
C. Slope
D. Error

Answer: C. Slope — b is the change in trend per time unit.

7. Coding time values is mainly done to:

A. Increase sample size
B. Simplify calculations
C. Eliminate trend
D. Reduce observations

Answer: B. Simplify calculations — Coding centres and simplifies time calculations.

8. The general second-degree trend equation is:

A. ■=a+bt

B. ■=a+bt+ct²

C. ■=ab■

D. ■=a/t

Answer: B. ■=a+bt+ct² — A quadratic trend includes a squared time term.

9. In a second-degree trend, coefficient c determines the:

A. Seasonal effect

B. Curvature

C. Random error

D. Time interval

Answer: B. Curvature — The quadratic coefficient controls curvature.

10. Percent of trend is calculated as:

A. Trend/Actual × 100

B. Actual/Trend × 100

C. Actual–Trend

D. Trend–Actual

Answer: B. Actual/Trend × 100 — Percent of trend compares actual value with trend value.

11. A commonly used method to measure seasonal variation is:

A. Least squares method

B. Ratio-to-moving-average method

C. Regression only

D. Chi-square test

Answer: B. Ratio-to-moving-average method — The ratio-to-moving-average method estimates seasonal indices.

12. A seasonal index of 100 indicates demand is:

A. Below average

B. Above average

C. Equal to average

D. Zero

Answer: C. Equal to average — 100 represents the average seasonal level.

13. A seasonal index greater than 100 indicates the observed value is:

A. Below average

B. Equal to average

C. Above average

D. Negative

Answer: C. Above average — Values above 100 indicate above-average seasonal activity.

14. Increased umbrella sales during every rainy season indicate:

A. Trend

B. Cyclical variation

C. Seasonal variation

D. Irregular variation

Answer: C. Seasonal variation — A recurring within-year pattern is seasonal.

15. A sudden decline caused by an earthquake is:

A. Trend

B. Seasonal variation

C. Cyclical variation

D. Irregular variation

Answer: D. Irregular variation — Unexpected events create irregular variation.

16. The commonly used method to determine a trend line is:

A. Chi-square test
B. Least squares method
C. ANOVA
D. Binomial distribution

Answer: B. Least squares method — Least squares fits a trend equation.

17. The completely unpredictable component of a time series is:

A. Trend
B. Seasonal variation
C. Cyclical variation
D. Irregular variation

Answer: D. Irregular variation — Irregular variation is random and unpredictable.

18. For an odd number of equally spaced observations, the coded value of the middle year is:

A. –1
B. 0
C. 1
D. 2

Answer: B. 0 — The middle time point is assigned zero.

19. Time series data are generally collected at:

A. Random intervals
B. Regular intervals
C. Unequal intervals
D. Different locations

Answer: B. Regular intervals — Time series observations are ordered at regular time intervals.

20. A seasonalized value is obtained by:

A. Deseasonalized value × seasonal index/100
B. Actual/Trend
C. Trend–Actual
D. Seasonal index/100 only

Answer: A. Deseasonalized value × seasonal index/100 — The seasonal index is applied proportionally to the trend or base value.